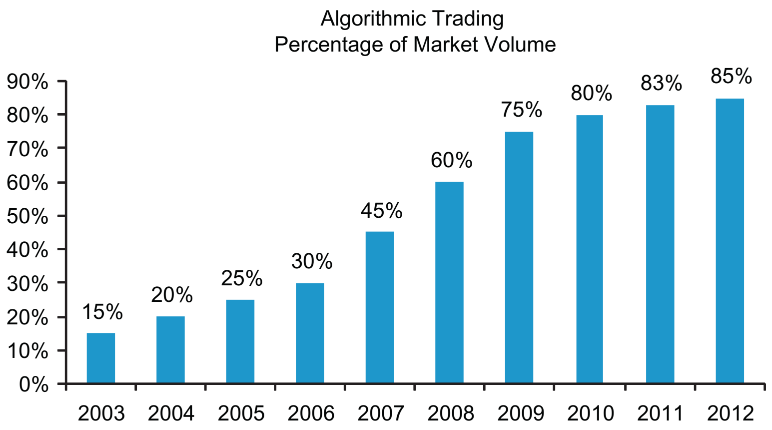

■ **Figure 1.9** Algorithmic Trading Percentage of Market Volume.

day, as well as the changing market environment (discussed above). These trends are shown in Figure 1.9. Over the years there have been various sources providing algorithmic trading estimates. For example, Tabb Group and Aite Group have published participation rates for buy-side algorithmic trading usage that are lower than our figures reported in this book. Our estimates include the execution's end product. So, even if the investor did not trade directly with an algorithm but did in fact route the algorithm to a broker who ultimately transacted the shares with an algorithm, those shares are included with the algorithmic trading volume figures.

The decade 2000−2010 was also associated with changing investor styles and market participants. We analyzed market participant order flow by several different categories of investors: traditional asset managers (including mutual funds, indexers, quantitative funds, and pension funds), retail investors, hedge funds (including statistical arbitrage and proprietary trading funds), market makers, and high frequency traders. In our definition, the high frequency trader only consisted of those investors considered liquid or rebate traders. We discuss the different types of high frequency trading below.

In 2003−2004 market volumes were led by asset managers, accounting for 40% of total volume. High frequency traders had almost negligible percentages in 2003 but grew to about 10% of the total market volumes in 2006. During the financial crisis, high frequency/rebate traders accounted for about 33% of volumes followed by hedge funds (21%). The biggest change we have observed over 2000−2012 is the decrease in asset manager volumes from 40% (2003) to about 23% (2012), and the increase in high frequency trading from 1−3% to about 30% of total volumes.